2. DATA COLLECTION AND PREPROCESSING PHASE

2.1 Raw Data Sources

Source

Kaggle – Online Payments Fraud Detection Dataset

Dataset Size

6,362,620 rows

Format

CSV

File Size

470 MB

2.2 Data Quality Report

Missing Values

No missing values detected.

Duplicate Records

No duplicates found.

Class Imbalance

Class	Count
Legit	6,354,407
Fraud	8,213

Fraud ratio ≈ 0.13%

2.3 Data Exploration and Preprocessing

Dropped Columns

- nameOrig
- nameDest
- isFlaggedFraud

Selected Features

- step
- type
- amount
- oldbalanceOrg
- newbalanceOrig
- oldbalanceDest
- newbalanceDest

Encoding

```
{'CASH_IN':0, 'CASH_OUT':1, 'DEBIT':2, 'PAYMENT':3, 'TRANSFER':4}
```

Scaling

StandardScaler applied.